Windows HDD File Recovery Suite

User Manual

A practical guide to scanning, reviewing, and restoring files safely

For Windows 10 and 11 | NTFS and FAT32 hard disk drives

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Before scanning: stop using the affected drive and prepare a separate physical disk for recovered files.

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Important: The provided screenshots show simulation mode. No real drive is read in that mode.

1. About this manual

This manual explains how to set up and use Windows HDD File Recovery Suite to look for deleted files, review recovery candidates, and save selected files to a safe location. It is written for students and other non-technical users. Follow the safety steps before starting: recovery results depend on whether deleted data has already been overwritten. This edition reflects the GitHub repository main branch at commit 3eae31e (12 October 2026).

> **Scope**
>
> The project targets Windows 10/11 x64 and NTFS or FAT32 hard disk drives. It can run with a local web interface or a hosted web interface connected to a Recovery Agent running on the users Windows PC. The supplied screenshots show simulation mode; mock data does not read a physical disk. Real scans require the Recovery Agent and real recovery engine.

The suite has a web interface backed by a recovery service and Windows scanning engine. In local mode, the app is normally at http://localhost:3000. In hosted mode, the website is a control panel and the Recovery Agent on the users PC performs scans and restores locally through http://127.0.0.1:8000.

2. Safety before recovery

> **Stop using the affected drive**
>
> Do not save, download, install, or copy files to the drive that contained the deleted data. New writes can overwrite recoverable content.

1. **Choose a separate destination drive.** Recovered files must be saved to a different physical disk from the source. Do not choose another partition on the same physical disk.
2. **Protect the Windows system drive.** Windows continues writing to its system disk. For the best chance, shut down and connect the affected disk to another computer as a secondary drive.
3. **Avoid formatting or repair prompts.** If Windows asks to format a disk, cancel the prompt. Do not run repair tools before attempting recovery.
4. **Consider professional help for failing disks.** Clicking, repeated disconnects, or read errors may indicate physical failure. Stop scanning if the disk appears unstable.

The engine is designed to open source drives read-only and checks the restore destination before writing. These safeguards reduce risk but cannot guarantee that every file can be recovered.

3. Set up the application and Recovery Agent

1. **For the local project build, start the components.** In PowerShell at the project root, run scripts\build-engine.ps1. Optionally run scripts\setup-database.ps1 for PostgreSQL. Then run scripts\start-backend.ps1 (add -DatabaseUrl if configured) and scripts\start-web.ps1. Allow the backend elevation prompt. The site opens at http://localhost:3000.
2. **For a hosted website, open Agent setup.** Install Python 3.12 and select Add python.exe to PATH. Download recovery-agent.zip from Agent setup and extract it, for example to C:\RecoveryAgent.
3. **Start the hosted Recovery Agent.** Double-click start-agent.bat in the extracted folder and approve the Windows Administrator prompt. The first run installs components and needs internet. Keep its window open and wait for “Uvicorn running on http://127.0.0.1:8000”.
4. **Allow the browser connection if prompted.** When the browser asks whether the website may connect to apps or devices on your computer or local network, allow it. Agent setup should show connected and indicate whether the real engine is ready or simulation mode is active.
5. **Sign in and check the status.** Create an account if needed, then sign in. In the local project setup, the first account is administrator. A “Mock Data” or simulation indicator means no physical disks are being read. Use your own credentials; do not reuse credentials visible in sample images.

> **Permissions**
>
> If the dashboard says Administrator rights are required, close the backend and restart it from an elevated PowerShell or Command Prompt. Signing in to the web app does not grant raw-disk permissions to the backend.

The sign-in page lets a new user create an account. The built-in User Manual tab provides topic navigation and safety guidance before you scan.

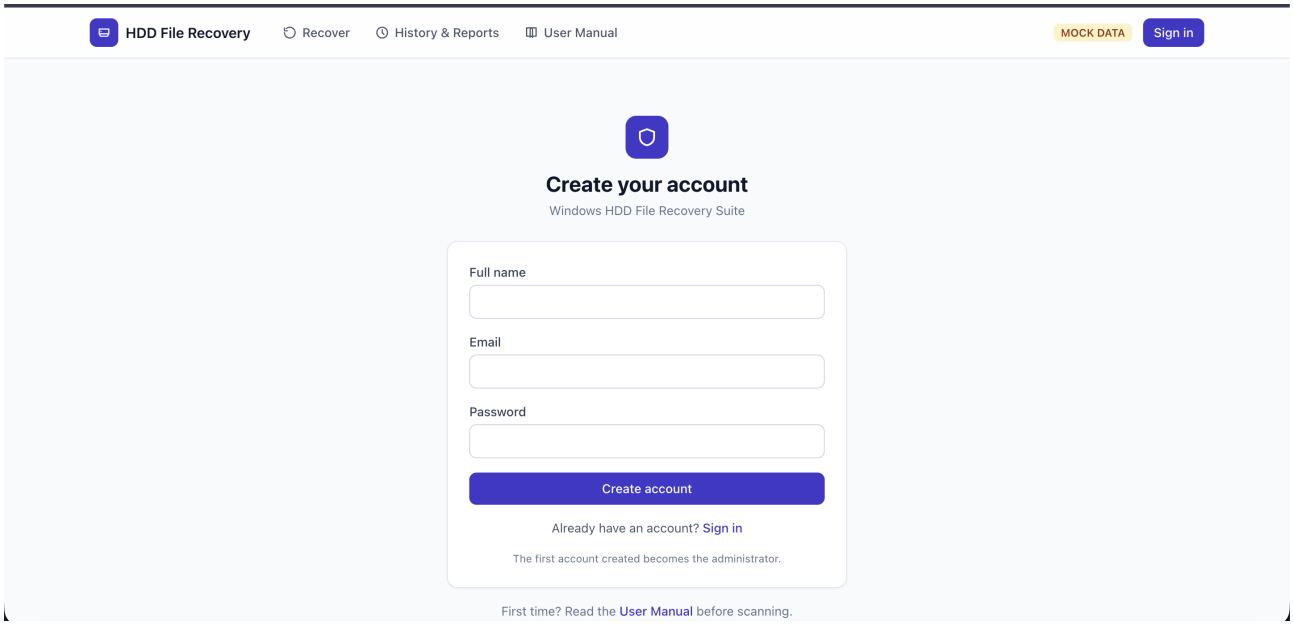


Figure 1. Account creation screen. The banner indicates mock mode.

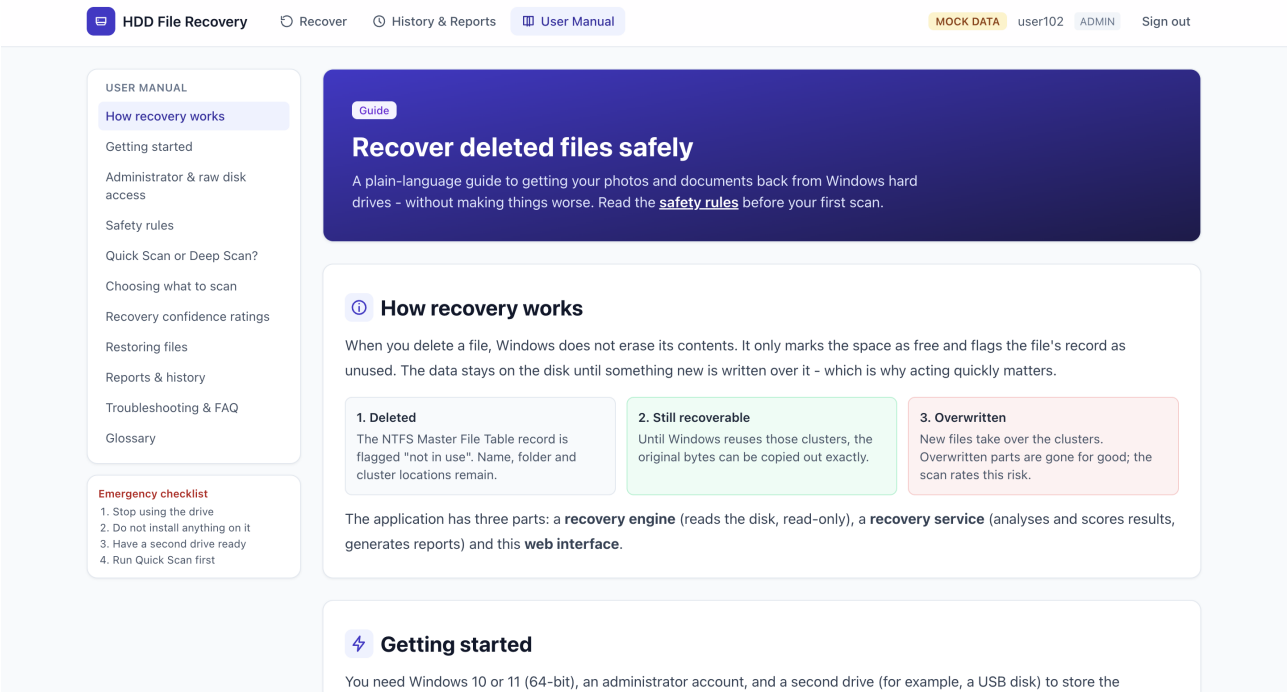


Figure 2. Built-in User Manual with topic navigation.

4. Choose a recovery scenario

On Recover, the first wizard step asks what happened. Selecting a scenario suggests a scan mode; you can change the mode later.

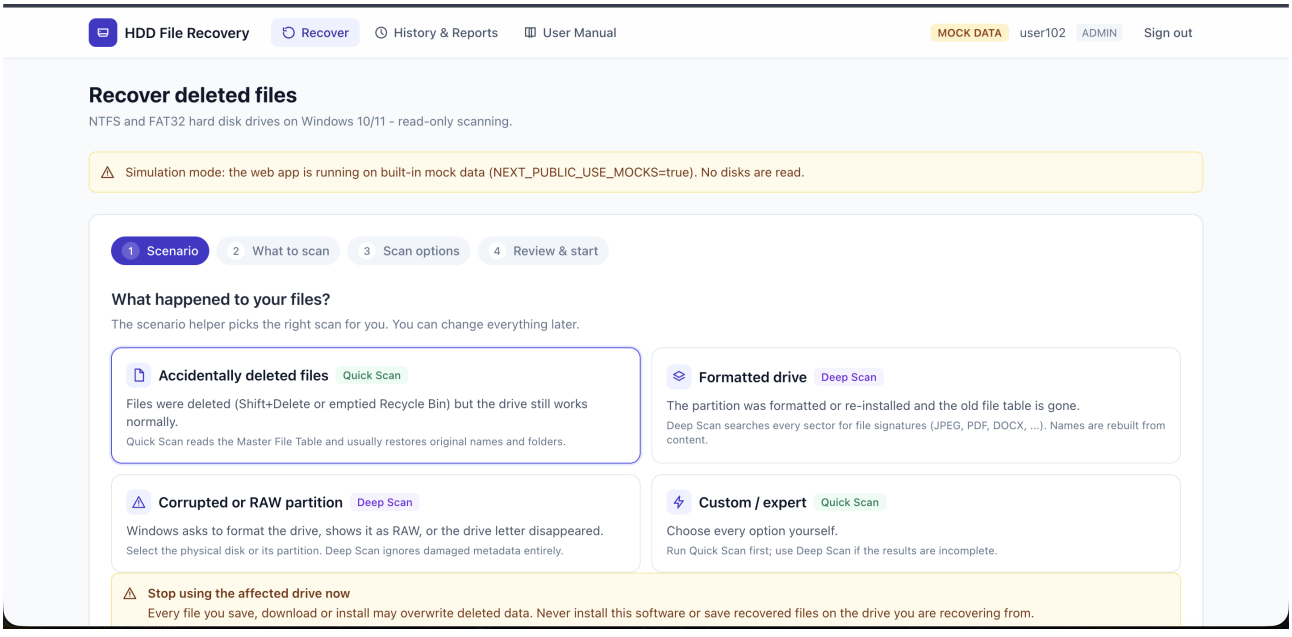


Figure 3. Choose the scenario that best describes the data loss.

Situation	Suggested starting point	What to expect
Deleted files; drive still works	Quick Scan	Usually retains original names and folders.

Situation	Suggested starting point	What to expect
Drive was formatted or reinstalled	Deep Scan	Searches sectors for file signatures; names may be reconstructed.
Partition is RAW or damaged	Deep Scan	Select the physical disk or affected partition.
Custom or uncertain case	Quick Scan first	Try Deep Scan if needed files are missing.

Scan mode

Quick Scan reads file-system metadata such as the NTFS Master File Table or FAT directories. Deep Scan looks for supported file signatures across sectors and can take much longer, especially on large disks.

5. Configure and start a scan

1. **Choose what to scan.** Select All drives, Single drive / partition, or Single folder. For a single target, carefully confirm the drive letter or physical disk using its size and label.
2. **Refresh drive detection if needed.** Use Refresh if a drive was connected after opening the page. If a drive is missing, check its cable and Windows Disk Management; do not initialize or format it.
3. **Select Quick Scan or Deep Scan.** Use the scenario recommendation as a starting point. Deep Scan may take hours on large disks.
4. **Optionally filter file types.** Choose extensions such as PDF, DOCX, JPG, or MP4. Leave every type unselected to search all supported types.
5. **Review advanced options.** Deep Scan can optionally limit carving to unallocated space. Use this only when you understand the trade-off; it may miss files outside the selected space.
6. **Review and acknowledge the safety notice.** Confirm the source, scan mode, and separate restore destination. Start the scan only after checking these details.

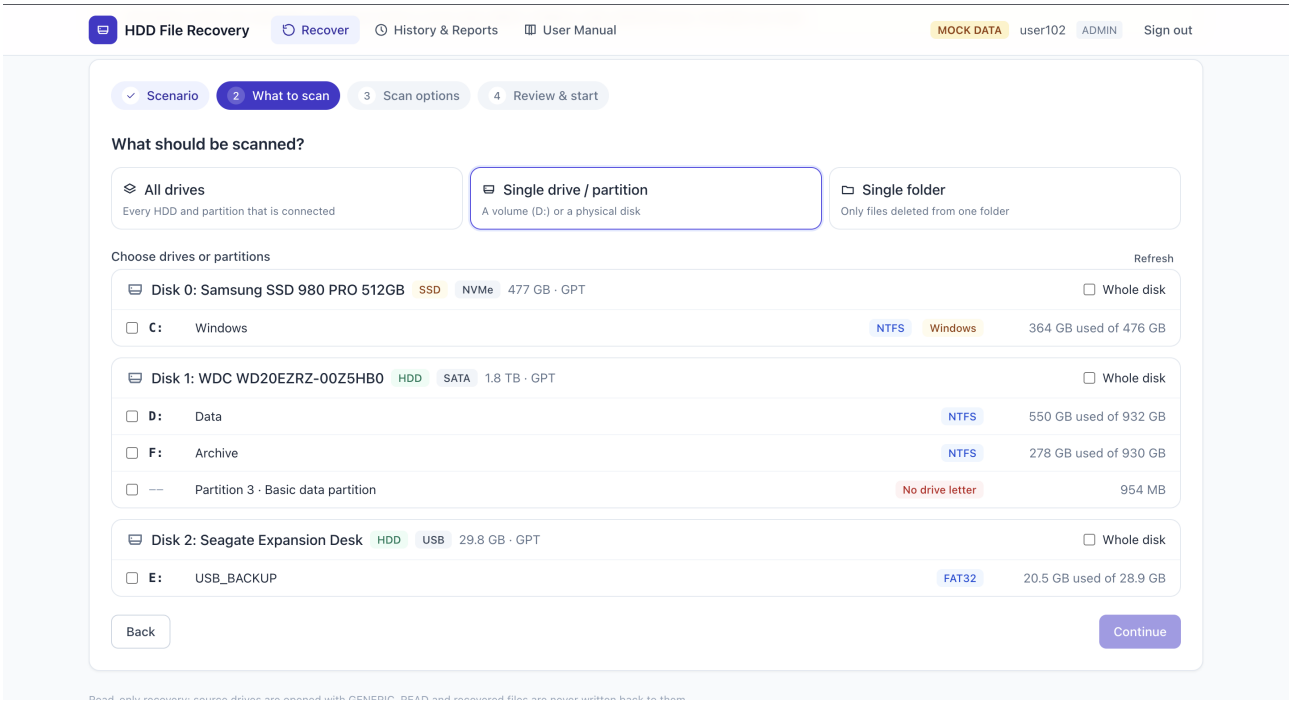


Figure 4. Choose the source drive or partition carefully. Example drive details are simulated.

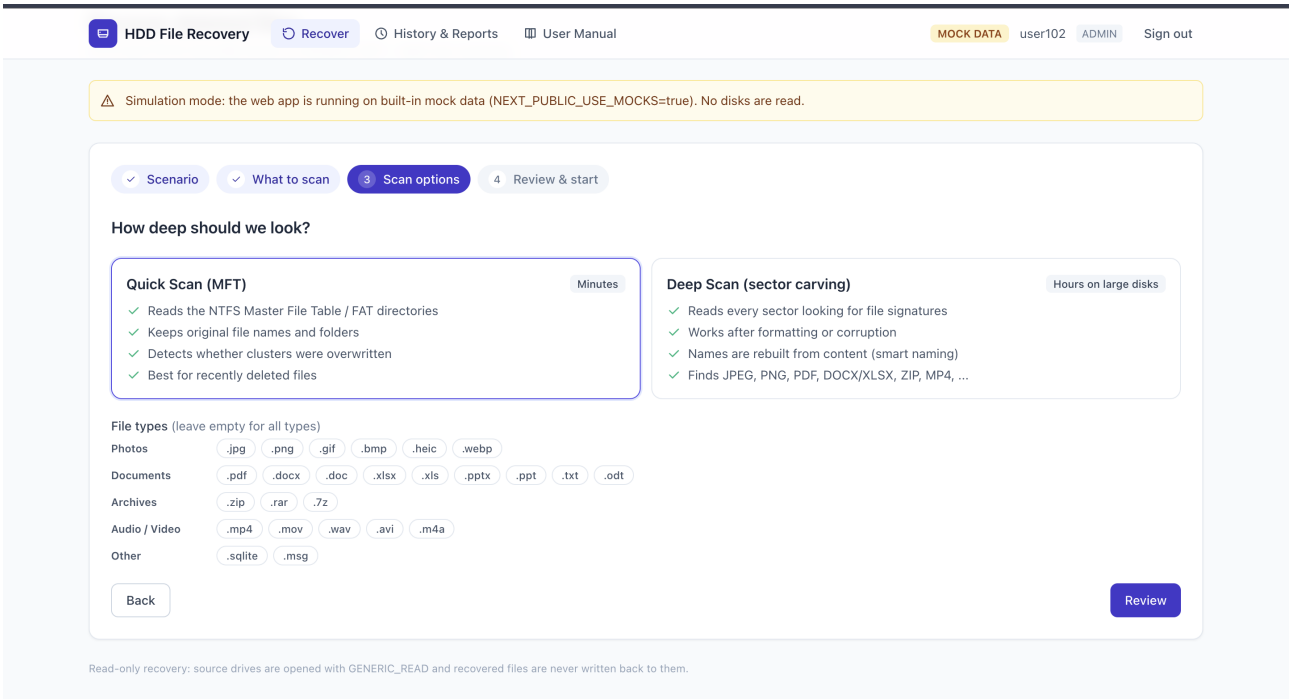


Figure 5. Select Quick Scan or Deep Scan and optional file type filters.

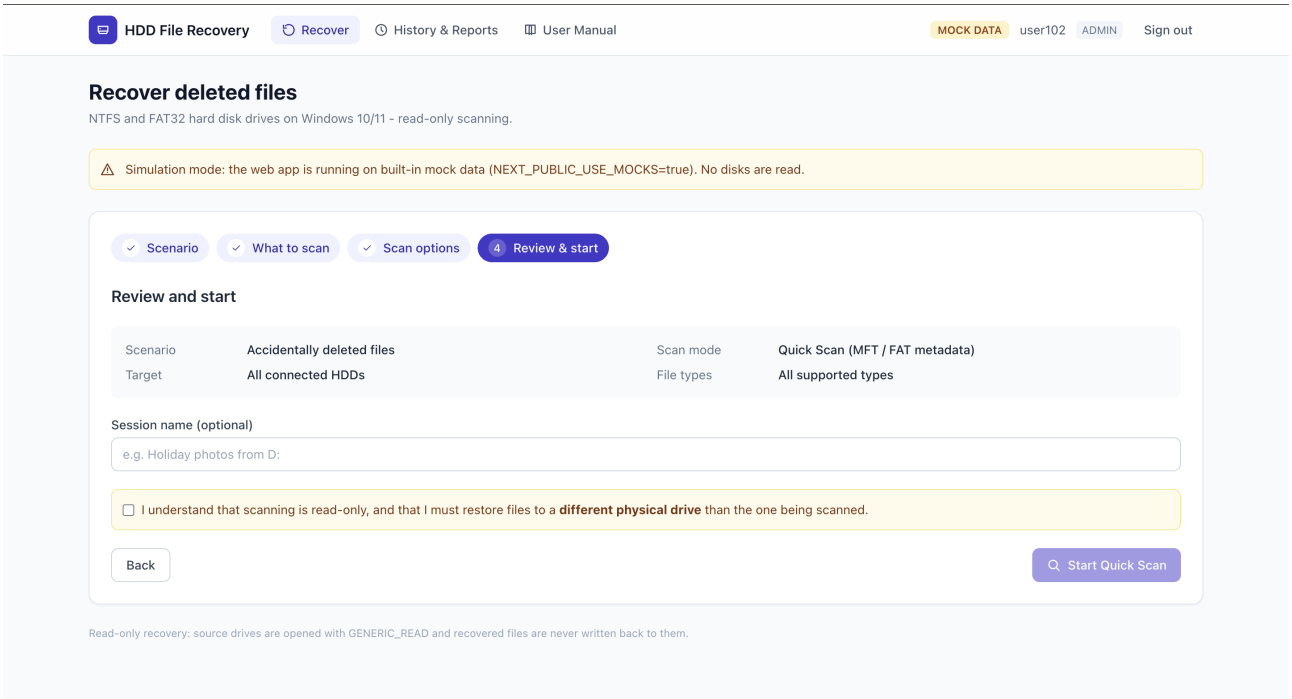


Figure 6. Review the scan scope and safety acknowledgement before starting.

SSD warning

The project is designed for HDD recovery. SSD TRIM may permanently clear deleted blocks. The interface may block SSD scanning unless you explicitly override the warning; overriding does not improve recovery odds.

6. Follow scan progress and review results

The recovery workspace displays progress, scan phase, speed or estimated time when available, and any read errors. Keep the computer powered on and avoid disconnecting the source drive. Use Cancel if you need to stop; a cancelled or partial scan may still have results.

When analysis completes, review the session statistics and result list. Use search and filters to narrow the candidates. Open a preview when available, and inspect the file name, type, size, path, and confidence rating before restoring.

Rating	Meaning	Suggested action
High	Metadata and content checks indicate a stronger candidate.	Preview and verify the file after restoring.
Medium	Some content or metadata may be incomplete.	Preview carefully; restore only if useful.
Low	The file may be fragmented, overwritten, or have estimated size.	Expect partial or damaged content.

A confidence rating is an estimate, not a guarantee. Deep Scan may reconstruct a name from content, and files found by carving may be incomplete if the original file was fragmented.

7. Restore selected files

1. **Select only the files you want.** Use previews and confidence ratings to prioritize important candidates.
2. **Choose a separate physical destination.** Never restore onto the source disk, even if you choose a different partition or folder on that disk.
3. **Check the destination validation message.** The application checks the target volume and free space. If validation fails, select a different disk with enough available space.
4. **Start restore and wait for completion.** The service writes the selected files to the destination and records the restore outcome.
5. **Verify the recovered files.** Open copies from the destination and check that documents, photos, or videos are usable. Keep an additional backup of important recovered data.

> **No overwrite**
>
> The recovery engine is designed not to overwrite existing destination files. If a name already exists, follow the application's result message and choose a safe destination.

8. History, search, and reports

Open History & Reports from the navigation bar to revisit saved sessions. Open a session to inspect its results. Completed sessions can be searched by label, filtered, and used to generate a PDF report. The History page also provides document-content search when indexing is available.

Use Import candidates.json only for a candidates file produced for this application. Deleting a session removes its stored scan results and history entry; it does not remove files already restored to your destination. Confirm before deleting.

Generated reports can summarize scan statistics and findings. Store reports with care because they may contain file names, paths, and other information about recovered content.

9. Troubleshooting

Message or symptom	What to do
Recovery service not reachable	Start the backend service and check the address/configuration. Confirm it is listening locally.
Administrator rights required	Restart the backend from an elevated terminal, then refresh the app.
No drives detected	Reconnect the drive, refresh detection, and confirm Windows can see it without formatting.
SSD warning	The application is HDD-focused. SSD TRIM commonly makes deleted data unrecoverable.
No files found	Confirm the correct source and scope. Try Deep Scan after Quick Scan, if safe. Results are not guaranteed.
Preview fails or recovered file will not open	Try another candidate, check confidence, and restore to a different destination. The source data may be overwritten or fragmented.
Scan is very slow	Deep scans read sectors and can take hours. Keep the drive connected and the computer powered.

If the drive makes unusual noises or repeatedly disconnects, stop. Further scans can worsen a failing device.

10. Limitations and glossary

The updated repository includes a prebuilt Windows engine and regression checks on synthetic images, but it has not yet been validated against physical disks on Windows hardware. Raw device access, Windows IOCTLs, and UAC still need a first test on a real Windows PC; test with a spare USB HDD first. Recovery depends on the storage medium and whether deleted blocks have been reused. SSDs with TRIM, overwritten sectors, fragmentation during signature carving, encrypted files, and unsupported file systems can prevent complete recovery. EFS-encrypted files cannot be decrypted by this suite. The README notes that compressed NTFS files may be recovered as raw clusters and may have low confidence.

Term	Meaning
Source	The drive or folder being scanned.
Destination	A separate physical disk where recovered copies are saved.
MFT	NTFS Master File Table; records file names, locations, and metadata.
Carving	Finding file content by recognizable signatures when directory metadata is unavailable.

Term	Meaning
Confidence	An estimate based on metadata and content checks; not a promise of successful recovery.
Mock data	Simulation data used to demonstrate the interface without reading real disks.

For installation and service configuration, refer to the project README and setup scripts supplied with the software.